

--	--	--	--	--

Centre Number

--	--	--	--	--	--	--	--	--

Student Number

2021 YEAR 12

Advanced Mathematics

Task 2

Monday 1st March, 2021

Q	Marks
MC	
5	
6	
7	
8	
9	
10	
11	
12	
Total	

General Instructions:

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:
34

Section I - 4 marks

- Allow about 7 minutes for this section

Section II - 30 marks

- Allow about 48 minutes for this section

This question paper must not be removed from the examination room.

This assessment task constitutes 20% of the course.

BLANK PAGE

Do NOT write in this area.

Section I

4 marks

Allow about 7 minutes for this section

Use the multiple-choice sheet for Question 1–4

1 Find the value of $\log_4 93$, correct to 3 significant figures.

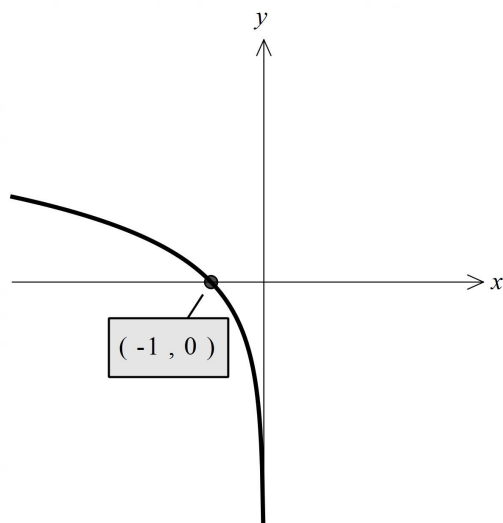
- (A) 1.97
- (B) 1.968
- (C) 3.27
- (D) 3.270

2 Find $\frac{d}{dx} \cos(x + 1)$

- (A) $\cos(x + 1)$
- (B) $\sin(x + 1)$
- (C) $-\sin(x + 1)$
- (D) $-\cos x \sin(x + 1)$

Examination continues on next page

- 3 Differentiate $e^x \sin x$ with respect to x .
- (A) $e^x \cos x$
- (B) $-e^x \cos x$
- (C) $e^x(\sin x + \cos x)$
- (D) $e^x(\sin x - \cos x)$
- 4 Which equation corresponds to the graph below?



- (A) $y = -\ln(x)$
- (B) $y = \ln(-x)$
- (C) $y = e^{-x} - 1$
- (D) $y = -e^{-x}$

End of Section I

Section II

30 marks

Allow about 48 minutes for this section

In Questions 5 – 12, your response should include relevant mathematical reasoning and/or calculations.

Question 5 (5 marks)

Differentiate the following, with respect to x , and fully simplify:

(a) $f(x) = 6x - 7 + e^{-2x}$ 1

.....

.....

.....

(b) $y = 3^{-x}$ 1

.....

.....

.....

.....

(c) $y = \log_e(\sec x)$ 3

.....

.....

.....

.....

.....

.....

.....

.....

Examination continues on next page

Question 6 (4 marks)

Consider the stem and leaf plot below with 17 scores.

0	58
1	04668999
2	00026
3	00

- (a) Give the 5-point summary.

2

.....

.....

.....

.....

.....

- (b) State the value of any outliers. Justify your answer with calculations.

2

.....

.....

.....

.....

.....

.....

.....

Do NOT write in this area.

Question 7 (4 marks)

Consider the function $f(x) = 3 - e^{-x}$

- (a) Evaluate $\lim_{x \rightarrow \infty} f(x)$

1

.....

.....

.....

- (b) Sketch the graph of $f(x)$, clearly stating the exact value of all intercepts.

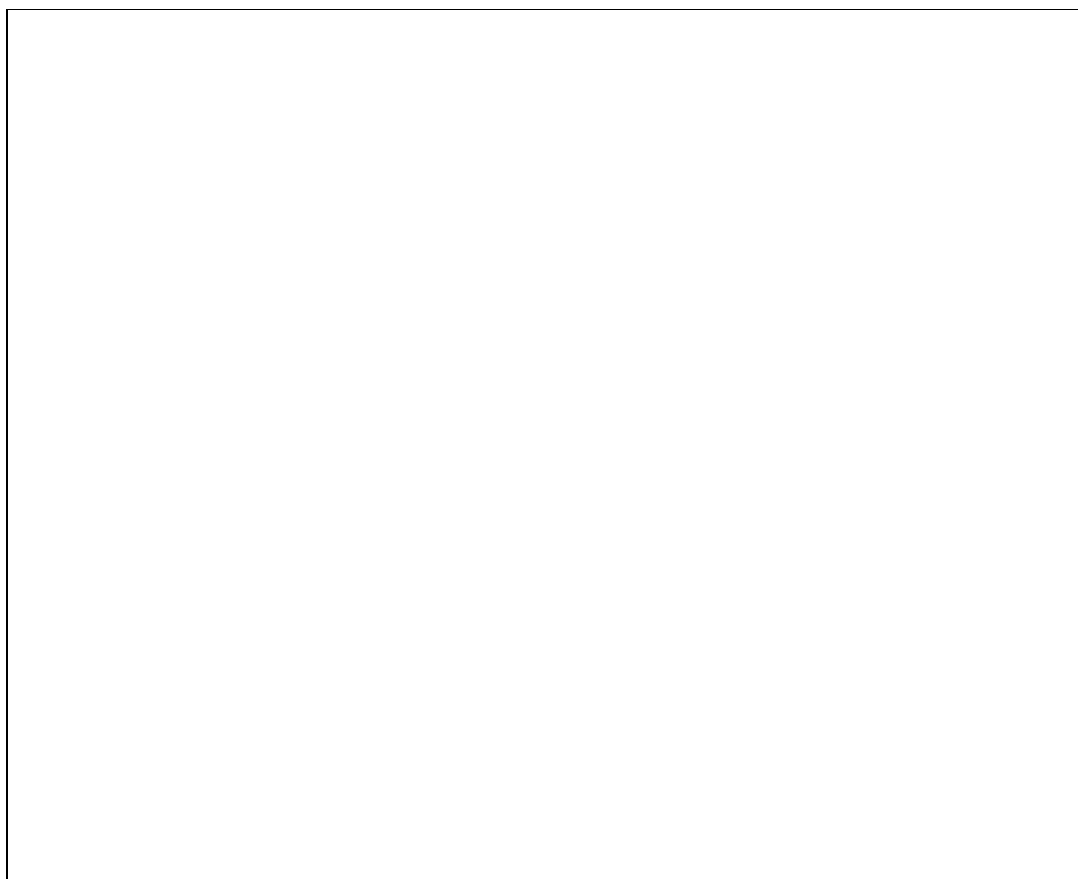
3

.....

.....

.....

.....



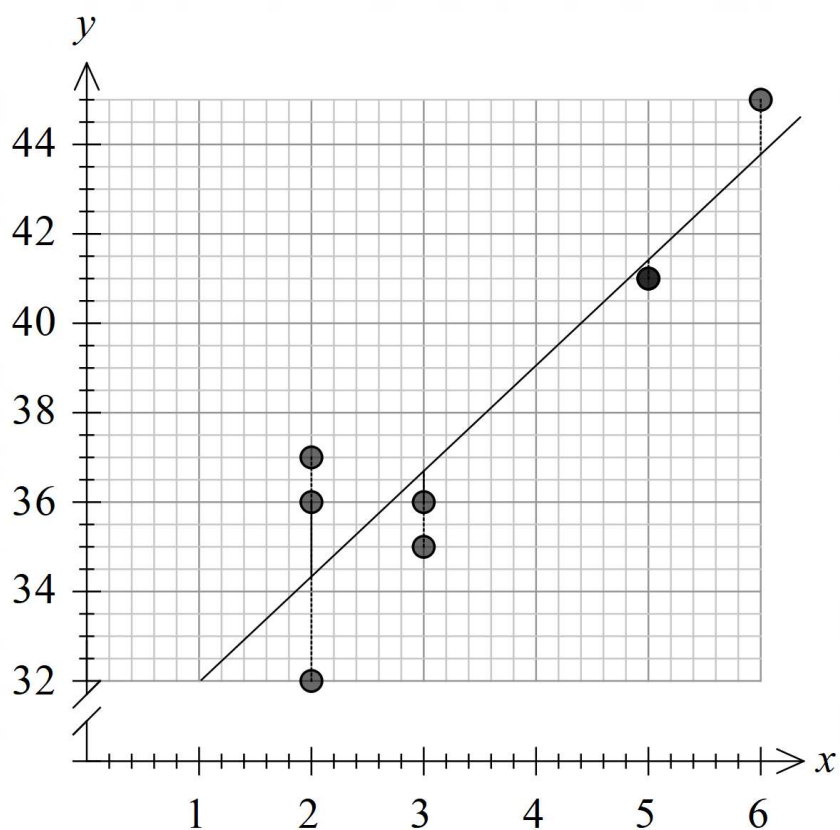
Examination continues on next page

Question 8 (4 marks)

A set of data was collected comparing the age of cars to their stopping distance during a brake test.

The collected data is shown in the table and graphed in the scatterplot shown.

Age of car (years)	2	2	5	3	6	2	5	3
Stopping Distance (m)	36	37	41	35	45	32	42	36



- (a) Find the Pearson correlation coefficient, r , correct to 3 decimal places.

1

.....

.....

.....

.....

- (b) Find the equation of the least-squares line of best fit, giving the coefficients to 1 decimal place. **1**

.....

.....

.....

.....

.....

.....

- (c) Use your result from part (b) to predict the stopping distance of car that is 4 years old. Comment on the reliability of your prediction. **2**

.....

.....

.....

.....

.....

.....

.....

.....

Examination continues on next page

Question 9 (2 marks)

If $\log_{10} x = 0.6$ and $\log_{10} y = 0.2$, evaluate $\log_{10} \left(\frac{x}{\sqrt{y}} \right)$.

2

.....

.....

.....

.....

.....

.....

.....

.....

Question 10 (2 marks)

Given that $f(x) = \tan^3 x$, calculate $f' \left(\frac{\pi}{4} \right)$.

2

.....

.....

.....

.....

.....

.....

.....

.....

Do NOT write in this area.

Question 11 (4 marks)

Super Mine World 64 is a video game known for "speedrunning", where players aim to complete the game in as short a run time as possible. A list of the world record fastest times are listed in the table below to the nearest second.

4

840s	848s	859s	912s	924s	931s	937s	939s	951s
------	------	------	------	------	------	------	------	------

When a new score was added to this list, the sample mean and interquartile range both decreased.

Using interval notation, state the interval in which this new score must lie.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do NOT write in this area.

Examination continues on next page

Question 12 (5 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula

$$E = k \times 10^{1.5R}$$

where R is the magnitude of the earthquake on the Richter scale and k is a constant. An earthquake of magnitude 3 radiates 2000 Megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(R - 1)$ **3**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Find the magnitude of an earthquake which radiates 5 million Megajoules of energy correct to 2 decimal places. **2**

.....

.....

.....

.....

.....

.....

.....

End of examination

Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Section I

4 marks

Allow about 7 minutes for this section

Use the multiple-choice sheet for Question 1–4

- 1 Find the value of $\log_4 93$, correct to 3 significant figures.

(A) 1.97

(B) 1.968

☒ (C) 3.27

(D) 3.270

- 2 Find $\frac{d}{dx} \cos(x + 1)$

(A) $\cos(x + 1)$

(B) $\sin(x + 1)$

☒ (C) $-\sin(x + 1)$

(D) $-\cos x \sin(x + 1)$

Examination continues on next page

3 Differentiate $e^x \sin x$ with respect to x .

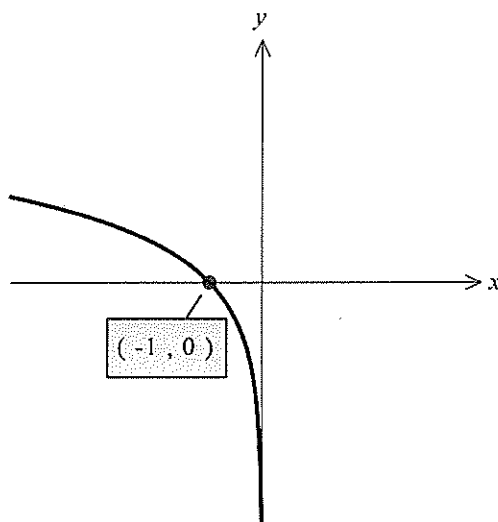
(A) $e^x \cos x$

(B) $-e^x \cos x$

(C) $e^x(\sin x + \cos x)$

(D) $e^x(\sin x - \cos x)$

4 Which equation corresponds to the graph below?



(A) $y = -\ln(x)$

(B) $y = \ln(-x)$

(C) $y = e^{-x} - 1$

(D) $y = -e^{-x}$

End of Section I

Section II

30 marks

Allow about 48 minutes for this section

In Questions 5 – 12, your response should include relevant mathematical reasoning and/or calculations.

Question 5 (5 marks)

Differentiate the following, with respect to x , and fully simplify:

(a) $f(x) = 6x - 7 + e^{-2x}$ 1

$$f'(x) = 6 - 2e^{-2x}$$

(b) $y = 3^{-x}$ 1

$$y' = -\ln 3 \times 3^{-x}$$

(c) $y = \log_e(\sec x)$ 3

$$\text{Let } u = \sec x$$

$$= (\cos x)^{-1}$$

$$u' = -(-\sin x)(\cos x)^{-1}$$

$$= \frac{\sin x}{\cos^2 x}$$

1 mark for derivative
of $\sec x$

$$y = \log_e u$$

$$y' = \frac{u'}{u}$$

$$= \frac{\frac{\sin x}{\cos^2 x} \times \cos x}{\frac{1}{\cos x}}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

1 mark for
simplifying

1 mark for correctly
applying

$$\frac{d \ln f(x)}{dx} = \frac{f'(x)}{f(x)}$$

Examination continues on next page

Question 6 (4 marks)

Consider the stem and leaf plot below with 17 scores.

0	58	Median
1	04668999	
2	00026	
3	00	

- (a) Give the 5-point summary.

$$\frac{n+1}{2} = \frac{17+1}{2} = 9^{\text{th}} \text{ score}$$

2

$$x_{\text{min}} = 5$$

$$Q_1 = 15$$

$$Q_2 = 19$$

$$Q_3 = 21$$

$$Q_4 = 30$$

-1 mark for any mistakes

- (b) State the value of any outliers. Justify your answer with calculations.

2

$$IQR = 21 - 15 = 6$$

$\therefore 5$ is an outlier as $5 < 6$

$$1.5 \times IQR = 9$$

$$\text{Upper fence} = 21 + 9 = 30$$

$$\text{Lower fence} = 15 - 9 = 6$$

1 mark for IQR

1 mark for stating outlier is less than the lower fence = 6

Do NOT write in this area.

Question 7 (4 marks)

Consider the function $f(x) = 3 - e^{-x}$

- (a) Evaluate $\lim_{x \rightarrow \infty} f(x)$

1

$$= 3 - 0$$

$$= 3$$

- (b) Sketch the graph of $f(x)$, clearly stating the exact value of all intercepts.

3

When $x = 0$

$$f(0) = 3 - e^0$$

$$= 3 - 1$$

$$= 2$$

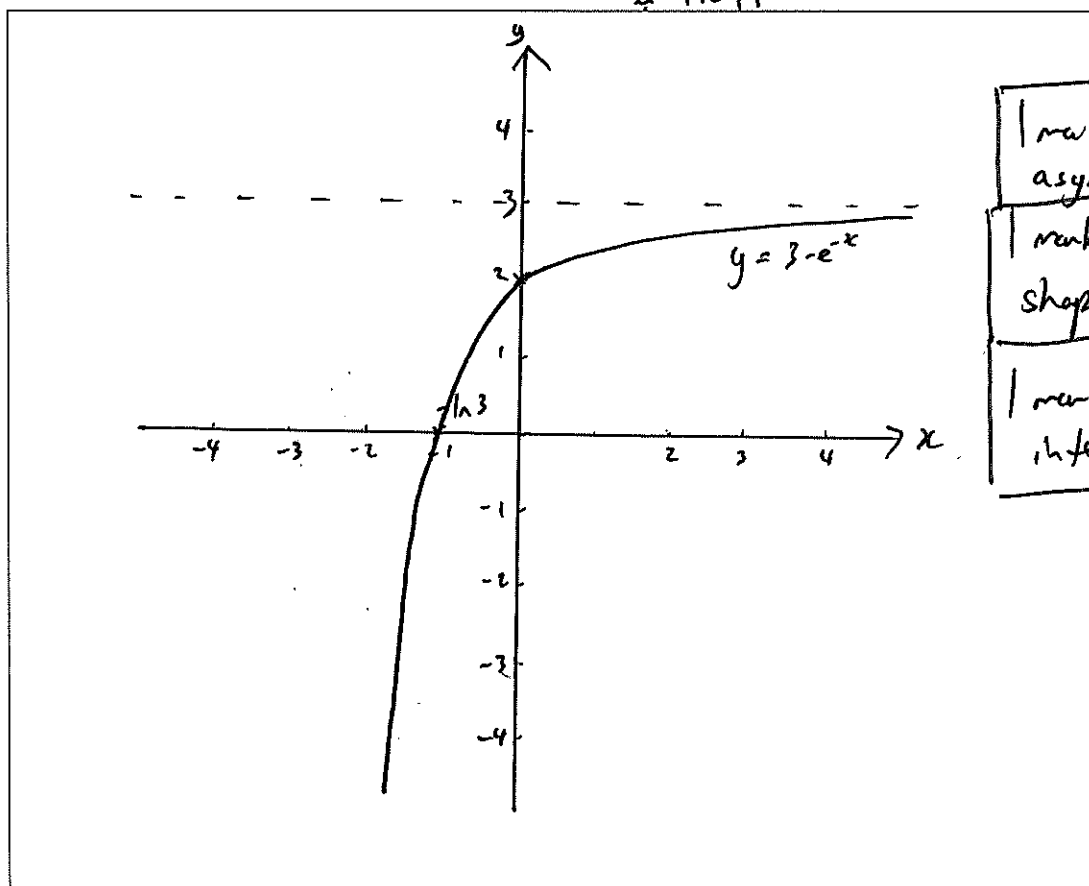
When $f(x) = 0$

$$3 - e^{-x} = 0$$

$$3 = e^{-x}$$

$$\ln 3 = -x$$

$$x = -\ln 3 \text{ or } x = \ln \frac{1}{3}$$



1 mark for asymptote

1 mark for shape

1 mark for both correct intercepts

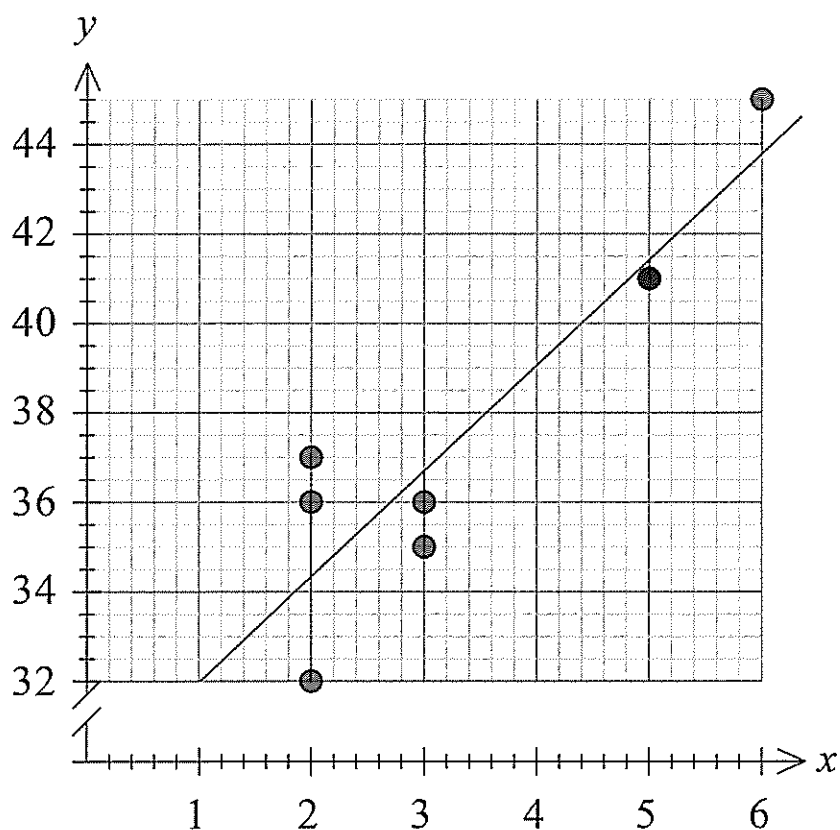
Examination continues on next page

Question 8 (4 marks)

A set of data was collected comparing the age of cars to their stopping distance during a brake test.

The collected data is shown in the table and graphed in the scatterplot shown.

Age of car (years)	2	2	5	3	6	2	5	3
Stopping Distance (m)	36	37	41	35	45	32	42	36



- (a) Find the Pearson correlation coefficient, r , correct to 3 decimal places.

1

$r = 0.917$
Using calculator

- 1 mark for incorrect rounding

- (b) Find the equation of the least-squares line of best fit, giving the coefficients to 1 decimal place. 1

$$A = 29.4$$

$$y = 29.4 + 2.4x$$

$$B = 2.4$$

or

Using calculator

Stopping distance = $29.4 + 2.4 \times \text{Age of car}$

- (c) Use your result from part (b) to predict the stopping distance of car that is 4 years old. Comment on the reliability of your prediction. 2

$$A + 4B = 39.2 \text{ (1 d.p.)}$$

Using calculator

1 mark (must not have used rounded values from (b))

Prediction is reliable as it is interpolating using the trend.

1 mark for reasoning "reliable" and "interpolation"

Examination continues on next page

Question 9 (2 marks)

If $\log_{10} x = 0.6$ and $\log_{10} y = 0.2$, evaluate $\log_{10} \left(\frac{x}{\sqrt{y}} \right)$.

2

$$\begin{aligned}\log_{10} \left(\frac{x}{\sqrt{y}} \right) &= \log_{10} x - \log_{10} \sqrt{y} \\ &= \log_{10} x - \log_{10} y^{\frac{1}{2}} \\ &= \log_{10} x - \frac{1}{2} \log_{10} y \\ &= 0.6 - \frac{1}{2} \times 0.2 \\ &= 0.5\end{aligned}$$

1 mark for separating two logs

1 mark for final result

Question 10 (2 marks)

Given that $f(x) = \tan^3 x$, calculate $f' \left(\frac{\pi}{4} \right)$.

2

$$f'(x) = 3 \tan^2 x \times \sec^2 x$$

1 mark for derivative

$$\begin{aligned}f' \left(\frac{\pi}{4} \right) &= 3 \times 1^2 \times (\sqrt{2})^2 \\ &= 6\end{aligned}$$

1 mark for final result

Do NOT write in this area.

Question 11 (4 marks)

Super Mine World 64 is a video game known for "speedrunning", where players aim to complete the game in as short a run time as possible. A list of the world record fastest times are listed in the table below to the nearest second.

4

840s	848s	859s	912s	924s	931s	937s	939s	951s
------	------	------	------	------	------	------	------	------

When a new score was added to this list, the sample mean and interquartile range both decreased.

Using interval notation, state the interval in which this new score must lie.

$$\bar{x} = 904.5$$

$$Q_1 = \frac{848 + 859}{2}$$

$$= 853.5$$

$$Q_3 = \frac{937 + 939}{2}$$

$$= 938$$

$$IQR = 84.5$$

Let y be the new score.

As $y \leq 904$ and the list is now 10 scores,

the new Q_3 becomes 937

$$937 - 84.5 = 852.5$$

$$\therefore 852.5 \leq y \leq 904.5$$

However, the list is quoted to the nearest second.

1 mark for recognising new $Q_3 = 937$

1 mark for $Q_3 - IQR$

1 mark for $\bar{x}$ and IQR

$$\therefore 853 \leq y \leq 904$$

or

$$[853, 904]$$

1 mark for either interval notation and rounded integers

Examination continues on next page

Question 12 (5 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula

$$E = k \times 10^{1.5R}$$

where R is the magnitude of the earthquake on the Richter scale and k is a constant. An earthquake of magnitude 3 radiates 2000 Megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(R - 1)$

3

$$\begin{aligned} 2000 &= k \times 10^{1.5 \times 3} \\ k &= \frac{2000}{10^{4.5}} \quad \boxed{\text{1 mark for finding } k \text{ in any form}} \\ &= 2 \times 10^3 \times 10^{-4.5} \\ &= 2 \times 10^{-1.5} \\ E &= 2 \times 10^{-1.5} \times 10^{1.5R} \\ &= 2 \times 10^{1.5R - 1.5} \\ &= 2 \times 10^{1.5(R-1)} \\ \log_{10} E &= \log_{10} (2 \times 10^{1.5(R-1)}) \quad \boxed{\text{1 mark for correctly taking } \log_{10} \text{ of both sides}} \\ &= \log_{10} 2 + \log_{10} 10^{1.5(R-1)} \\ &= \log_{10} 2 + 1.5(R-1) \quad \boxed{\text{1 mark for final result with correct working}} \end{aligned}$$

- (b) Find the magnitude of an earthquake which radiates 5 million Megajoules of energy correct to 2 decimal places.

2

$$\begin{aligned} \log_{10} E &= \log_{10} 2 + 1.5(R-1) \\ \log_{10} E - \log_{10} 2 &= 1.5(R-1) \\ \log_{10} \left(\frac{E}{2} \right) &= 1.5(R-1) \\ R-1 &= \frac{\log_{10} \left(\frac{E}{2} \right)}{1.5} \\ R &= \frac{\log_{10} \left(\frac{E}{2} \right)}{1.5} + 1 \end{aligned}$$

When $E = 5000000$

$$\begin{aligned} R &= \frac{\log_{10} \left(\frac{5000000}{2} \right)}{1.5} + 1 \\ &= 5.27 \text{ (2 d.p.)} \end{aligned}$$

$\boxed{\text{1 mark for substitution}}$
 $\boxed{\text{1 mark for correct final result.}}$

End of examination